

CURRICULUM VITAE

Hao, Shugang

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EDUCATION

- **Doctor of Philosophy** 08/2022
- *Engineering Systems and Design, Singapore University of Technology and Design, Singapore*
- **Bachelor of Engineering** 06/2017
- *Information Engineering, South China University of Technology, Guangzhou, Guangdong, China*

WORKING EXPERIENCE

- **Research Fellow, Principal Investigator** 03/2026 - now
- *SUTD Wireless Innovation Centre, Singapore University of Technology and Design, Singapore*
- **Postdoctoral Research Fellow** 09/2022 - 02/2026
- *Singapore University of Technology and Design, Singapore*

RESEARCH INTERESTS

- AI-native networks
- AI alignment
- Game theory and mechanism design

AWARDS

- **SUTD President's Graduate Fellowship**, 2017 - 2022.
- **ACM MobiHoc 2019/SIGMobile student travel grant**.

Research Grant

- **Distributed Autonomous Wi-Fi Algorithms for Channel Access and Sharing**
- *Principal Investigator, SGD 0.79 million, 03/2026 - now.*

TEACHING EXPERIENCES

- **Teaching Assistant**
ESD-40.012 Manufacturing and Service Operation, 2020 Spring & 2019 Spring,
Singapore University of Technology and Design.

PUBLICATIONS

- Submitted papers
 - (S3) **Shugang Hao** and Lingjie Duan, “*Prompt-Aware WiFi Scheduling for Generative AI Services*”, submitted to MobiHoc’26.
 - (S2) **Shugang Hao** and Lingjie Duan, “*Transformer-Based In-Context Learning for Unknown and Non-Concave Network Utility Maximization*”, submitted to MobiHoc’26.
 - (S1) **Shugang Hao** and Lingjie Duan, “*LLM Alignment against Strategic Preference-Data Poisoning: A Partial Information Disclosure Mechanism*”, submitted to IEEE TIFS.
- Journal papers
 - (J5) **Shugang Hao***, Hongbo Li* and Lingjie Duan, “*Theory of Transformer-Based In-Context Learning for Distributed Channel Access Optimization*”, IEEE Transactions on Mobile Computing (**IEEE TMC**), to appear, July 2026. (* The two authors contributed equally to this work.)
 - (J4) **Shugang Hao** and Lingjie Duan, “*Truthful Online Preference Aggregation for LLM Fine-Tuning in Mobile Crowdsourcing*,” IEEE Transactions on Mobile Computing (**IEEE TMC**), to appear, doi: 10.1109/TMC.2026.3696583, May 2026.
 - (J3) **Shugang Hao** and Lingjie Duan, “*To Save Mobile Crowdsourcing from Cheap-talk: A Game Theoretic Learning Approach*,” IEEE Transactions on Mobile Computing (**IEEE TMC**), vol. 23, no. 8, pp. 8418-8430, August 2024.
 - (J2) **Shugang Hao** and Lingjie Duan, “*To Help or Disturb: Introduction of Crowdsourced WiFi to 5G Networks*,” IEEE Transactions on Mobile Computing (**IEEE TMC**), vol. 22, no. 9, pp. 5583-5596, September 2023.
 - (J1) **Shugang Hao** and Lingjie Duan, “*Regulating Competition in Age of Information under Network Externalities*,” IEEE Journal on Selected Areas in Communications (**IEEE JSAC**), vol. 38, no. 4, pp. 697-710, April 2020.
- Conference papers
 - (C5) **Shugang Hao***, Hongbo Li* and Lingjie Duan, “*To Theoretically Understand Transformer-Based In-Context Learning for Optimizing CSMA*”, in ACM International Symposium on Mobile Ad Hoc Networking and Computing (**ACM MobiHoc**), Houston, Texas, USA, Oct. 27-30, 2025. (* The two authors contributed equally to this work.)
 - (C4) **Shugang Hao** and Lingjie Duan, “*Online Learning from Strategic Human Feedback in LLM Fine-Tuning*”, IEEE International Conference on Acoustics, Speech, and Signal Processing (**IEEE ICASSP**, oral), Hyderabad, India, April 6-11, 2025.
 - (C3) **Shugang Hao** and Lingjie Duan, “*Algorithm Design for Continual Learning in IoT Networks*”, IEEE International Conference on Acoustics, Speech, and Signal Processing (**IEEE ICASSP**), Hyderabad, India, April 6-11, 2025.
 - (C2) **Shugang Hao** and Lingjie Duan, “*To Save Crowdsourcing from Cheap-Talk: Strategic Learning from Biased Users*,” in the 21th International Symposium on Modeling and Optimization in Mobile, Ad hoc, and Wireless Networks (**IEEE WiOpt**), Singapore, August 24-27, 2023.
 - (C1) **Shugang Hao** and Lingjie Duan, “*Economics of Age of Information Management under Network Externalities*,” in ACM International Symposium on Mobile Ad Hoc Networking and Computing (**ACM MobiHoc**), Catania, Italy, July 2-5, 2019.

SERVICES

- **Journal Editorship**
 - Youngth Editor, High-Confidence Computing, July 2026.
 - Guest Editor, Special issue “Large Language Models for 6G Networks”, Intelligent and Converged Networks, 2026.
- **Conference Chairs**
 - TPC Co-Chair, IEEE ICC 2026 workshop on Near-Field Communications, Localization, and Sensing, Glasgow, Scotland, UK.
 - Web Chair, ACM SenSys 2024, Hangzhou.
 - Local Arrangement Chair, AIOISys 2024, Hangzhou.
 - Local Arrangement Chair, IEEE/IFIP WiOpt 2023, Singapore.
- **Technical Program Committee Member**
 - IEEE MSN 2025, Bandung, Indonesia.
 - AIOISys 2024, Hangzhou.
- **Technical Reviewers**
 - IEEE Journal on Selected Areas in Communications,
 - IEEE Transactions on Networking,
 - IEEE Transactions on Mobile Computing,
 - IEEE Internet of Things Journal,
 - IEEE Transactions on Wireless Communications,
 - ACM Transactions on Sensor Networks,
 - IEEE Transactions on Communications,
 - IEEE Transactions on Vehicular Technology,
 - IEEE Transactions on Cognitive Communications and Networking,
 - IEEE Communication Letters,
 - IEEE Wireless Communications Magazine,
 - ACM MobiHoc, IEEE INFOCOM, IEEE SECON, IEEE ICC, IEEE GLOBECOM, IJCNN.

INVITED TALKS

- **“The Economics of Competition under Network Externalities,”**
 - at Sun Yat-sen University, Guangzhou, Guangdong, China, 14/09/2023.
- **“To Help or Disturb: Introduction of Crowdsourced WiFi to 5G Networks,”**
 - at South China University of Technology, Guangzhou, Guangdong, China, 12/09/2023,
- **“To Save Crowdsourcing from Cheap-talk: Strategic Learning from Biased Users,”**
 - at Sun Yat-sen University (Zhuhai Campus), Zhuhai, Guangdong, China, 09/09/2023.
- **“Economics of Age of Information Management under Network Externalities,”**
 - at Guangdong University of Technology, Guangzhou, Guangdong, China, 31/08/2019,
 - at Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences, Shenzhen, Guangdong, China, 30/08/2019,
 - at South China University of Technology, Guangzhou, Guangdong, China, 27/08/2019.